CST8248 – Emerging Technologies

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Week 06

CST8248

Agenda

- More Demo
- Assignment 3
- House Keeping
  - Cluster Use
  - Thick provisioned disks
  - VM’s in ISO Storage
- Veeam

Veeam

- Backup and Recovery Solution
  - Physical Servers
  - Virtual Servers
  - Virtual Server Infrastructure
- https://www.veeam.com/veeam-free-online-training.html?ad=vbr
- Can integrate with Active Directory to allow management of users and groups

Veeam

- Backups can be managed by the Veeam Server or by the Veeam agent
  - Veeam Server (agentless) Backups allow target machines to be automatically backed up by the Veeam server from the Veeam server.
  - Veeam Server Backups are best for always on workloads
  - Veeam Agent Backups allow the agent machine to contact the Veeam server and send backup data. This is good for systems that are not always on or may be affect by metered connections

Veeam

- Types of Backups
  - Active Full,
  - Synthetic Full
  - Forever Forward Incremental
  - Forward Incremental
  - Reverse Incremental
- File Types
  - Full *.VBK (both active and synthetic)
  - Incremental
  - Forward incremental *.VIB
  - Reverse incremental *.VRB
  - Metadata *.VBM

Veeam

- Backup Process (Server Backup - Agentless)
  - Veeam Server will contact vCenter and query VM's in the backup job using vmtools.
  - A snapshot of the VM is taken to avoid system interuption
  - If this is a Full backup all files will be gathered
  - If this is an incremental backup only files that have changed since the last backup are gathered.
  - The files are transferred, compressed and stored in the backup repository(ies)
  - The snapshot is removed.

Veeam

- Backup Process (Server Backup w/Agent)
  - Veeam Server will contact the destination Machine using stored credentials.
  - The agent software is requested to gather and transmit the requisite files
  - If this is a Full backup all files will be gathered
  - If this is an incremental backup only files that have changed since the last backup are gathered.
  - The files are transferred, compressed and stored in the backup repository(ies)

Veeam

- Backup Process (Agent only)
  - The Veeam Agent will start the backup operation on schedule or immediately after a missed backup cycle
  - The agent software gathers and transmits the requisite files. Using the same method as before.
  - The Veeam server is contacted and a connection is established
  - The files are transferred, compressed and stored in the backup repository(ies)

Veeam - Types of Backups

- Active Full
  - All files a collected a normal full backup is performed
- Synthetic Full

- Forward Incremental
- Reverse Incremental

Veeam - Types of Backups

- Forever Forward Incremental

- Forever Forward Incremental Backup Method
- The forever forward incremental backup method produces a backup chain that consists of the first full backup file (.VBK) and a set of forward incremental backup files (.VIB) following it.
- During the first session of a backup job, Veeam Backup & Replication creates a full backup file on the backup repository. During subsequent backup job sessions, Veeam Backup & Replication copies only VM data blocks that have changed since the last backup job session and saves these blocks as an incremental backup file in the backup chain.
- After adding a new restore point to the backup chain, Veeam Backup & Replication checks the retention policy for the job and deletes outdated restore points.

Forward Incremental Backup Method

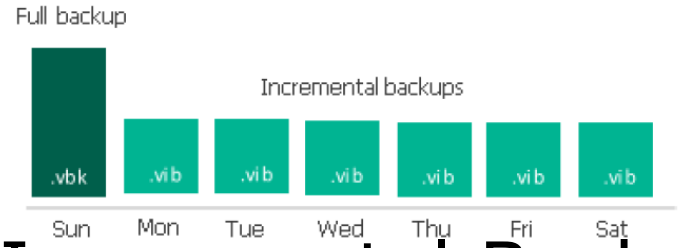
- Additionally, the forward incremental backup chain contains full backup files that "split" the backup chain into shorter series. The subsequent full backup files can be the following:
  - Active
    - The active full backup contains the copy of a VM. This backup is similar to the full backup created when you run a job for the first time. Veeam Backup & Replication retrieves data for the whole VM from the source, compresses and deduplicates it and stores it to the active full backup file.
  - Synthetic
    - The synthetic full backup also contains the copy of a VM. However, this copy is created from the backup files that you already have on the backup repository. Veeam Backup & Replication does not retrieve VM data from the source datastore.
- During the first backup job session, Veeam Backup & Replication creates a full backup file. During subsequent backup job sessions, Veeam Backup & Replication copies only VM data blocks that have changed since the last backup job session and saves these blocks as an incremental backup file in the backup chain. On a day when the synthetic or active full backup is scheduled, Veeam Backup & Replication creates a full backup file and adds it to the backup chain.
- After adding a new restore point to the backup chain, Veeam Backup & Replication checks the retention policy for the job and deletes outdated restore points.

Reverse Incremental Backup Method

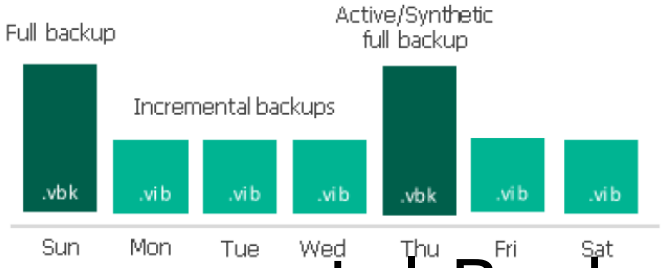
- The reverse incremental backup method produces a backup chain that consists of the last full backup file (.VBK) and a set of reverse incremental backup files (.VRB) preceding it.
- During the first backup job session, Veeam Backup & Replication creates a full backup file on the backup repository. During subsequent backup job sessions, Veeam Backup & Replication copies only VM data blocks that have changed since the last backup job session. Veeam Backup & Replication “injects” copied data blocks into the full backup file to rebuild it to the most recent state of the VM. Additionally, Veeam Backup & Replication saves the changed block in the reverse incremental backup file and places this file before the full backup file.
- After adding a new restore point to the backup chain, Veeam Backup & Replication checks the retention policy and deletes outdated restore points. For details, see Reverse Incremental Backup Retention Policy in the Veeam Backup & Replication User Guide.

Comparing Backup Chains

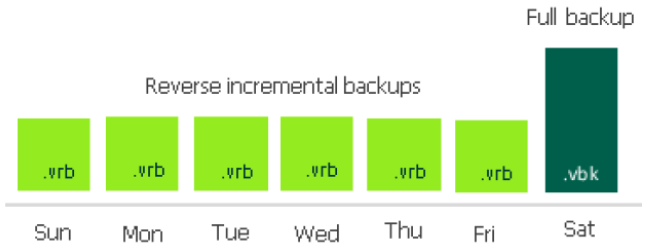
- Forever Forward Incremental Backup Method



- Forward Incremental Backup Method



- Reverse Incremental Backup Method



Developing an Effective Backup Strategy

- A proper Backup Strategy Consists of:
  - Scheduled Full Backups with Multiple Incremental or Differential Backups
  - Completed Automatically and reliably
  - Tested Regularly for consistency and viability

Developing an Effective Backup Strategy

- 3-2-1 Backup Schema
  - 3 Copies of the Data
  - 2 Types of Media
  - 1 Remote Storage Location
- Copies of Data
  - Typically, 3 copies is recommended.
  - Backup versioning is also recommended
    - Grandfather, Father, Son (these names will likely change)

Developing an Effective Backup Strategy

- Types of Media
  - Magnetic (Hard Disk, Tape)
  - Solid State (SSD, SD Card/other portable flash, USB Thumb drive)
  - Optical (CD, DVD, Blu-Ray)
  - Others?

Developing an Effective Backup Strategy

- Types of Locations
  - In Production
    - Fastest to restore, great for File History or recovering accidental deletion
    - Vulnerable to attack
  - On-Premise
    - Moderate time to restore
    - Potentially more resilient to attack

Developing an Effective Backup Strategy

- Types of Locations
  - Remote
    - Cloud Storage
    - Companion Site (ie. Head Office, Satellite Office)

    - Slow to restore
    - Most resilient to attack
    - More costly

Windows Admin Center

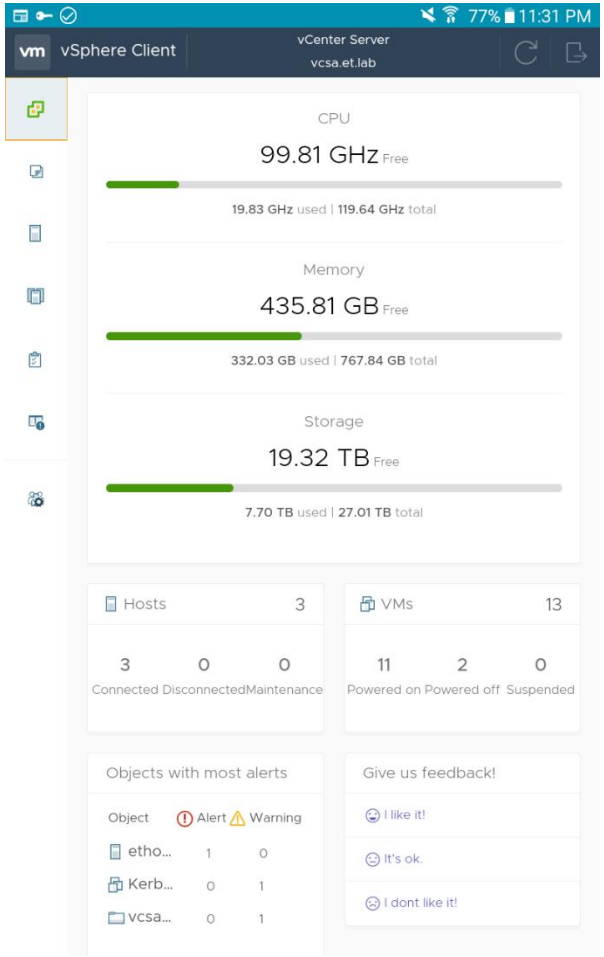
- Centralized Administration Platform
- Requires Windows Management Instrumentation (WMI)
- Integration with Azure for Hybrid cloud operations
- Requires Windows Server 2019

- Formerly known as project Honolulu

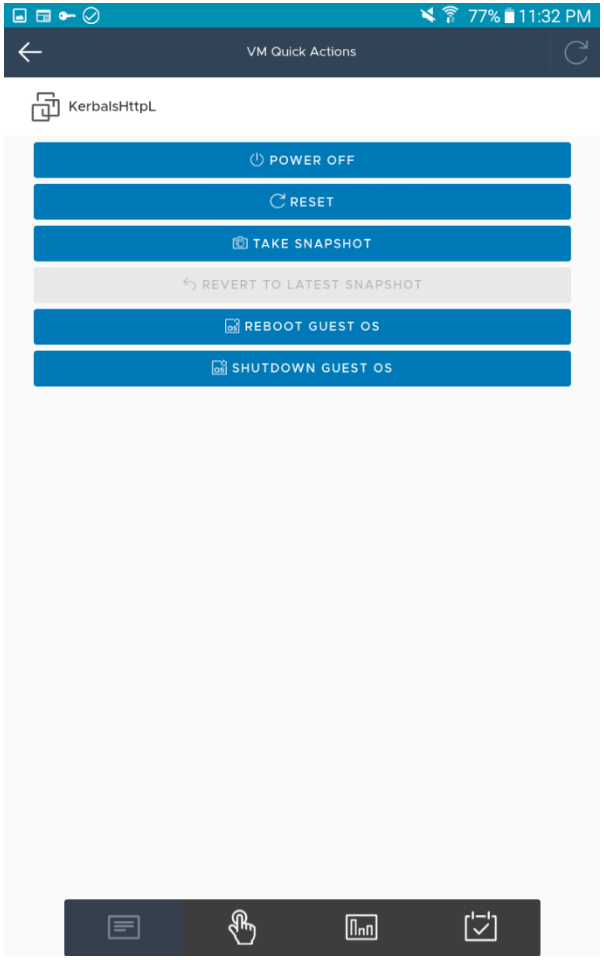
VMWare: vSphere Mobile

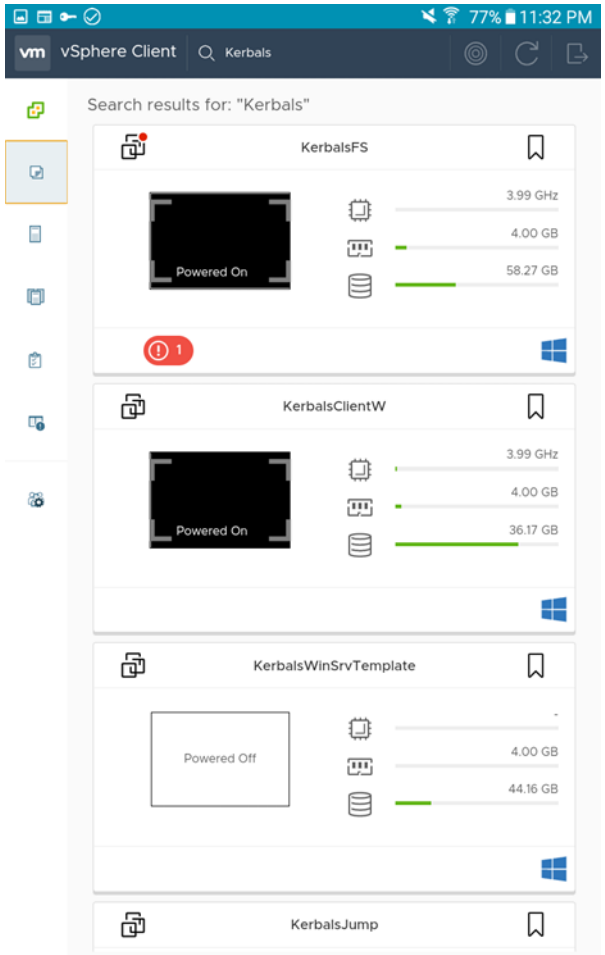
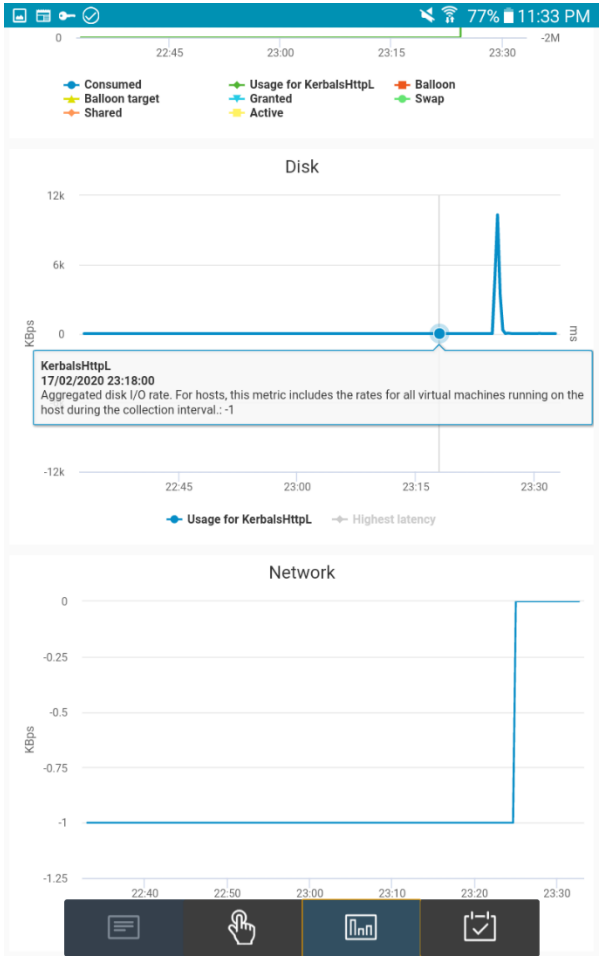
- Early Access application for Android
- Allows for Remote access to vCenter
- Requires port 443
- Currently free of charge in the google play store.

VMWare: vSphere Mobile



VMWare: vSphere Mobile





VMWare: vSphere Mobile